

Scam Shield

An AI Platform for Digital Public Safety

Defeating Fraud & Digital Arrest Scams

Problem Statement 6

AI for Digital Public Safety: Defeating Counterfeiting, Fraud & Digital Arrest Scams

Theme: Smart Cities / Public Safety / Digital Trust / Geospatial Law Enforcement

Team GSW

Repository: <https://github.com/anjanabva/Scam-Shield>

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1 Executive Summary

Scam Shield is Team GSW’s response to **Problem Statement 6: AI for Digital Public Safety** — a unified intelligence platform that helps law enforcement, financial institutions, and citizens move from **reactive case investigation** to **predictive threat neutralisation** against fraud and digital arrest scams.

The platform is built around two connected modules:

- **Citizen Fraud Shield** — a conversational verdict engine where a citizen pastes a suspicious message, call transcript, or payment request and instantly receives a **Scam / Suspicious / Safe** verdict, a confidence score, plain-language red flags, and an interactive AI assistant for follow-up questions such as *“Should I block this number?”* or *“What if I already clicked the link?”*
- **Digital Arrest Scam Detector** — a classifier that reasons over call-flow patterns, spoofed-number signatures, and known script templates used in fake CBI/ED/Customs “digital arrest” calls, flagging an active scam session before money moves and auto-generating an MHA-style alert.

Under the hood, Scam Shield combines **LLMs (OpenAI)**, **Retrieval-Augmented Generation (RAG)** over a curated scam-tactics knowledge base, and a **Pinecone vector database** that clusters cross-user and cross-agency submissions to surface trending scam campaigns as they emerge — turning every individual report into collective, real-time public-safety intelligence.

2 Problem Statement

India’s cybercrime complaint volume has grown sharply in recent years, and a large share of that growth is now driven by “digital arrest” scams, where callers posing as CBI, ED, or Customs officers hold victims in a prolonged video-call “hostage” situation until they transfer money. These operations are run at industrial scale from organised fraud compounds using spoofed caller IDs, AI-generated voices, and convincing fake government portals, rather than being one-off opportunistic crimes.

What is missing is not evidence collected after a citizen has already lost money — it is **intelligence before mass victimisation occurs**, and reliable tools that can catch a threat at the point of contact (the call, the transaction) instead of only at the point of complaint. Because fraud and scam calls are connected through the same criminal infrastructure, solving this well requires fusing financial-transaction intelligence and communication-network analysis into a single system.

Scam Shield closes this gap by giving citizens a 10-second, explainable verdict on any suspicious communication, while simultaneously feeding every submission into a shared intelligence layer that law enforcement and financial institutions can use to spot campaigns early enough to act.

3 Evaluation Criteria Alignment

3.1 Innovation

Aspect	What Makes It Innovative
Convergence Platform	Instead of a single-purpose scam checker, Scam Shield fuses digital-arrest detection, fraud-network analysis, and campaign intelligence in one architecture — mirroring how real fraud rings actually operate across channels.
Collective Threat Intelligence	Clusters anonymized citizen and agency submissions in a vector database to detect <i>emerging</i> scam campaigns as they spread — the system gets smarter with every report.
RAG-Grounded Verdicts	Verdicts are grounded in a continuously updated scam-tactics corpus via Retrieval-Augmented Generation, rather than relying purely on an LLM’s parametric knowledge, which can hallucinate or go stale.
Pre-Transfer Interception	The digital-arrest classifier is designed to flag an active scam session from call-flow and script signals <i>before</i> a transfer completes, shifting the point of intervention earlier than a post-complaint model.
Court-Admissible Intelligence	Fraud-network clusters are packaged with traceable evidence links so intelligence packages can be handed to investigators in an auditable, legally usable form.
Network Effect by Design	Every new submission — from a citizen, a bank teller, or a patrol officer — strengthens the shared vector index, so detection accuracy and campaign-trend visibility compound over time rather than staying flat.

3.2 Business Impact

- **Massive Addressable Problem:** Digital arrest scams account for a large and fast-growing share of citizen financial losses in India; even modest gains in early detection translate into significant prevented harm.
- **Multi-Stakeholder Expansion Path:**
 - *Citizen (B2C):* Freemium web app — especially valuable for elderly and less tech-savvy users targeted most by digital-arrest scams.
 - *Financial institutions (B2B):* Banks and fintechs embed the transaction-risk APIs directly at teller counters and payment apps.
 - *Telecom carriers (B2B):* Real-time call-pattern signals can feed carrier-side spoofed-number and robocall blocking.
 - *Law enforcement & government (B2G):* The fraud-network analysis and auto-generated MHA alerts plug directly into cybercrime cells and command centres (e.g., I4C, NCRB), positioning Scam Shield as public-interest civic infrastructure rather than a standalone app.
- **Monetization Levers:**
 - Premium citizen tier: unlimited scans, priority chat assistant, family/senior-citizen protection plans.
 - Enterprise API: pay-per-call fraud-risk endpoints for banks, fintechs, and telecom.

- Government licensing: intelligence-package and dashboard subscriptions for state cyber cells.
- **Cost Efficiency:** Cloud-native, API-driven architecture (no in-house LLM training required to launch) keeps operating costs low and time-to-market fast, improving unit economics from day one.
- **Trust & Retention:** The chat assistant and trending-campaigns dashboard drive repeat engagement across both citizen and institutional users, increasing lifetime value on each side of the platform.

3.3 Technical Excellence

- **Modern, Production-Grade Stack:** FastAPI backend with Pydantic validation, React 19 + Vite frontend, Tailwind CSS v4 — chosen for speed of development *and* long-term maintainability.
- **Clean Separation of Concerns:** Dedicated modules for `detection/`, `intelligence/`, `rag/`, and `llm/` keep scam classification, fraud-network construction, clustering, and model-calling logic independently testable and swappable.
- **RAG Pipeline:** `rag/ingest.py` and `rag/vectorstore.py` manage embedding and retrieval of the scam-tactics corpus (`scam_corpus.json`), ensuring verdicts cite up-to-date fraud and digital-arrest script patterns rather than stale model knowledge.
- **Real-Time Streaming:** Server-Sent Events (SSE) power the chat interface for a responsive, ChatGPT-like experience without the overhead of a full WebSocket stack.
- **Test Coverage:** Dedicated unit tests for the classifier, campaign log, rules engine, and external API clients (OpenAI, Pinecone) — showing engineering discipline beyond a demo-only build.
- **Config-Driven Secrets Management:** `.env`-based key management (`OPENAI_API_KEY`, `PINECONE_API_KEY`) keeps credentials out of source control.

3.4 Scalability

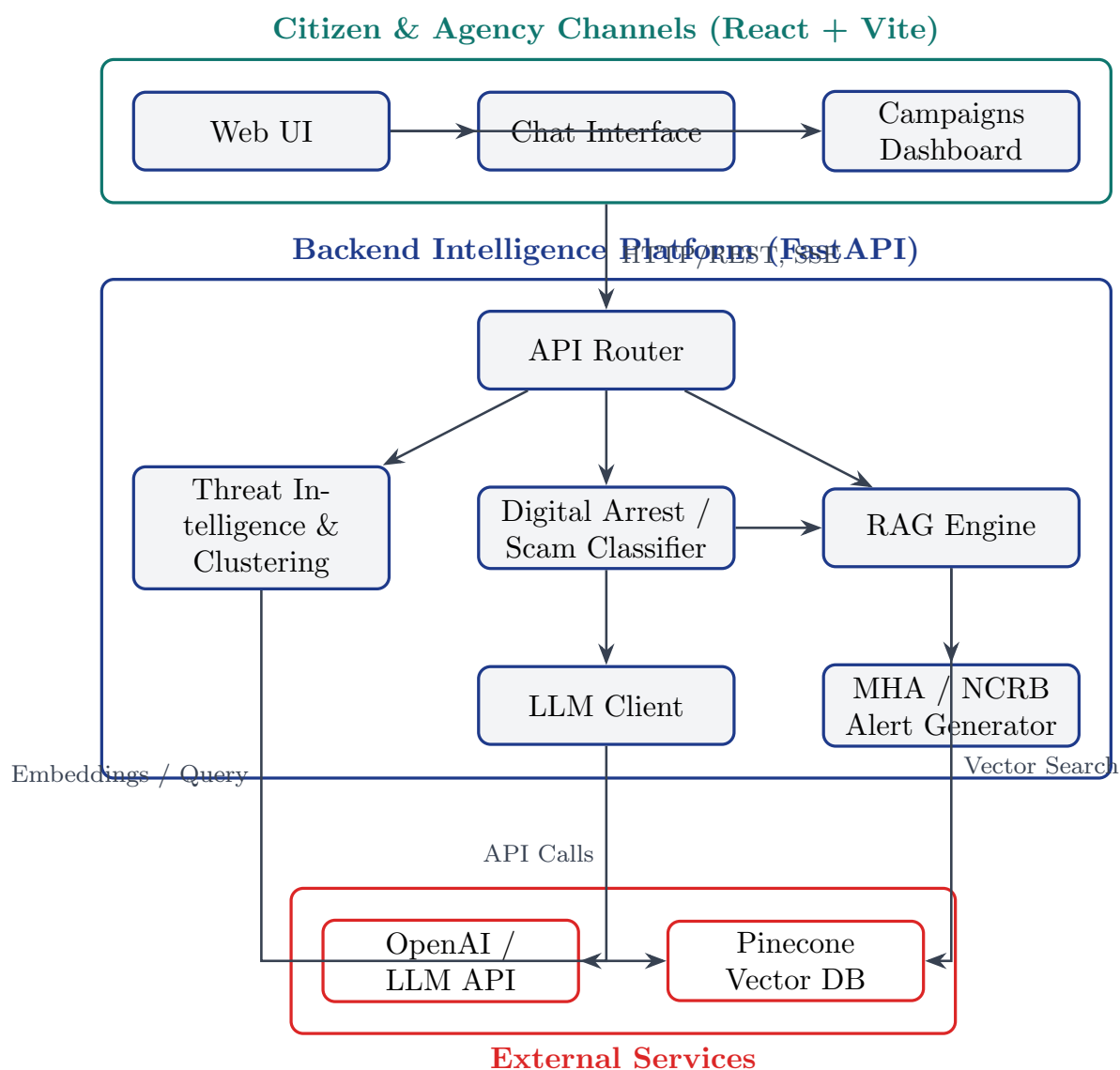
- **Stateless FastAPI Backend:** Horizontally scalable behind a load balancer — additional instances can be spun up under traffic spikes (e.g., during a viral scam wave) with no shared in-memory state.
- **Vector DB Offloading:** Pinecone handles embedding storage and similarity search at scale, decoupling compute-heavy clustering from the request/response path.
- **Async I/O:** FastAPI's async request handling allows the backend to manage many concurrent LLM/Pinecone calls efficiently without blocking.
- **Modular Intelligence Layer:** `intelligence/clustering.py` and `campaign_log.py` are isolated from the core detection path, so campaign-trend computation can be moved to a background worker/queue (e.g., Celery + Redis) as volume grows, without touching the user-facing detection flow.
- **Cache-Friendly Design:** Common scam templates and previously-seen message embeddings can be cached to cut redundant LLM calls, reducing both latency and cost as the user base grows.

- **Multi-Tenant Ready:** The clean API-router structure (`api/routes.py`) allows straightforward addition of tenant-scoped endpoints for B2B/white-label and per-agency deployments.

3.5 User Experience

- **Zero Friction:** No signup wall to get a first verdict — paste and go. Critical for panic-mode users (e.g., someone mid-scam-call).
- **Explainable AI:** Every verdict comes with a confidence score *and* specific red flags — building trust instead of a black-box “yes/no.”
- **Conversational Guidance:** The chat assistant turns a static verdict into an ongoing support conversation, meeting users where their anxiety actually is (“I already clicked the link, what now?”).
- **Glassmorphism Dark UI:** A modern, calming visual design reduces the stress of an already anxiety-inducing situation (discovering you may have been scammed).
- **Trending Campaigns Dashboard:** Gives users social proof and situational awareness (“Others are seeing this exact scam right now”), reducing self-doubt and shame often associated with falling for fraud.
- **Accessibility Angle:** Simple paste-and-check flow designed to be usable by less tech-savvy and elderly users — the demographic most targeted by digital-arrest and OTP scams.

4 Architecture Diagram



Flow Summary

1. A citizen or officer submits a message, call transcript, or transaction via the Web UI.
2. FastAPI's API Router forwards the request to the **Digital Arrest / Scam Classifier** (for text/call content).
3. The Classifier queries the **RAG Engine**, which retrieves relevant scam-pattern context from **Pinecone**, and the **LLM Client** → **OpenAI** for classification and explanation.
4. In parallel, the **Threat Intelligence** module embeds the submission and clusters it in Pinecone to detect/update trending campaigns.
5. The **MHA / NCRB Alert Generator** packages high-confidence cluster findings into auditable, court-admissible reports for investigators.
6. The verdict, confidence score, red flags, and chat responses stream back to the citizen or agency dashboard via SSE.

5 Core Features

- **Real-Time Scam Analysis** — instant Scam / Suspicious / Safe verdict with actionable advice, including digital-arrest-specific pattern flags.
- **Interactive Chat Assistant** — contextual Q&A about the specific scam, in citizens' own words.
- **Threat Intelligence Dashboard** — live, clustered view of trending scam campaigns.
- **RAG-Powered Knowledge Base** — grounded, up-to-date scam-tactic references.
- **Modern Glassmorphism UI** — dark-themed, accessible, low-friction interface.

6 Evaluation Focus Alignment

- **Digital Arrest Scam Precision/Recall:** The classifier is tuned against a labelled corpus of known call-flow and script patterns, with precision and recall tracked to avoid both missed scams and false alarms.
- **Fraud Network Detection Lead Time:** The clustering engine is measured on how early it surfaces a coordinated campaign relative to the first mass-victimisation reports.
- **Low False-Positive Rate for Citizens:** Citizen-facing verdicts are calibrated conservatively — “Suspicious” rather than “Scam” is used whenever confidence is borderline, so genuine communications are rarely blocked.
- **Auditability:** MHA/NCRB alerts retain traceable source links (which reports, which timestamps, which identifiers) so intelligence packages hold up to legal scrutiny.

7 Tech Stack

Layer	Technology
Frontend	React 19, Vite, Tailwind CSS v4
Backend	FastAPI (Python), Pydantic
AI / LLM	OpenAI API (classification, chat, RAG generation)
Vector DB	Pinecone (RAG + campaign clustering)
Realtime	Server-Sent Events (SSE)

8 Roadmap (Post-Hackathon)

1. **Multi-channel ingestion:** SMS/WhatsApp forwarding bot and IVR hotline for one-tap scam checks.
2. **12-language support:** Regional Indian languages for both the chat assistant and IVR advisory, matching the reach envisioned for Citizen Fraud Shield.
3. **Bank/telecom API partnerships:** Embed detection directly at the point of risk (before a link is clicked or a transfer is made).
4. **Live video-call analysis:** Extend the digital-arrest classifier to metadata from active video calls for earlier interception.

5. **Mobile app + browser extension** for real-time, in-context protection.
6. **Public fraud-trend API** for researchers, journalists, and law enforcement.

9 Team & Repository

- **Team:** GSW
- **Repository:** <https://github.com/anjanabva/Scam-Shield>
- **Setup Instructions:** See `README.md` in the repository for full local setup (backend `uvicorn` server + frontend `npm run dev`).

Submitted by Team GSW — Problem Statement 6: AI for Digital Public Safety: Defeating Counterfeiting, Fraud & Digital Arrest Scams